Raspberry Pi Hardware Preparation and Installation

Hardware Guide: For getting started with raspberry pi for the first time you will require the following hardware

1. Raspberry Pi (latest Model)
2. Monitor or TV
3. HDMI cable
4. Ethernet cable
5. USB keyboard
6. USB mouse
7. Micro USB power supply
8. 8GB or larger microSD card
9. SD Card Reader

Raspberry Pi 3 Model B:
The Raspberry Pi 3 is the third generation Raspberry Pi

Monitor or TV:
A monitor or TV with HDMI in can be used as a display with a Raspberry Pi

HDMI to HDMI Cable:
Connect Raspberry Pi to a Monitor or TV with a HDMI to HDMI cable.

Ethernet cable:
Ethernet cable will allow your Pi to connect with the internet

USB Keyboard and Mouse:
Any standard USB keyboard and mouse can be used with the Raspberry Pi

Power Supply:
It is recommended that you use a 5V, 2A USB power s upply for all models of Raspberry Pi.

SD Card:
The latest version of Raspbian, the default operating system recommended for the Raspberry Pi, requires an 8GB (or larger) micro SD card. SD card will store the operating systems as well as all the file and applications created by you.

1. Install Raspbian OS on microSD Card

1. Download: Download the Raspbian OS image file.
2. Extract: Unzip the downloaded file to get the disc image.
3. Format: Format the microSD card using SD Formatter.
4. Write: Use Win32 Disk Imager to write the OS image onto the SD card.
5. Ready: The SD card is now ready for the first-time setup.

2. Plugging in Raspberry Pi

1. Insert the microSD card into the Raspberry Pi.
2. Connect the keyboard and mouse to USB ports.
3. Turn on the monitor and select the correct input.
4. Connect the Raspberry Pi to the monitor using an HDMI cable.
5. Connect the internet using Ethernet or Wi-Fi.
6. Connect the micro-USB power supply to boot the Raspberry Pi.

GPIO: Light the LED with Python

Hardware Guide:
1. LED
2. Resistor
3. Connecting wires
4. Breadboard

Wiring up your Circuit:
1. Connect the GPIO22 (i.e. Physical Pin 15) Pin od raspberry pi to one end of the resistor.
2. Connect another end of resistor to the positive end (anode) of LED
3. Connect the negative end (cathode) of LED to Ground of raspberry pi.
4. Then Power on your raspberry pi

Software Guide (Short Notes)

1. Raspbian OS comes with preinstalled programming software.
2. We use Python for coding.
3. Open Menu → Programming → Python 3 (IDLE).
4. Click File → New File to start writing code.

```
import RPi.GPIO as GPIO
from time import sleep

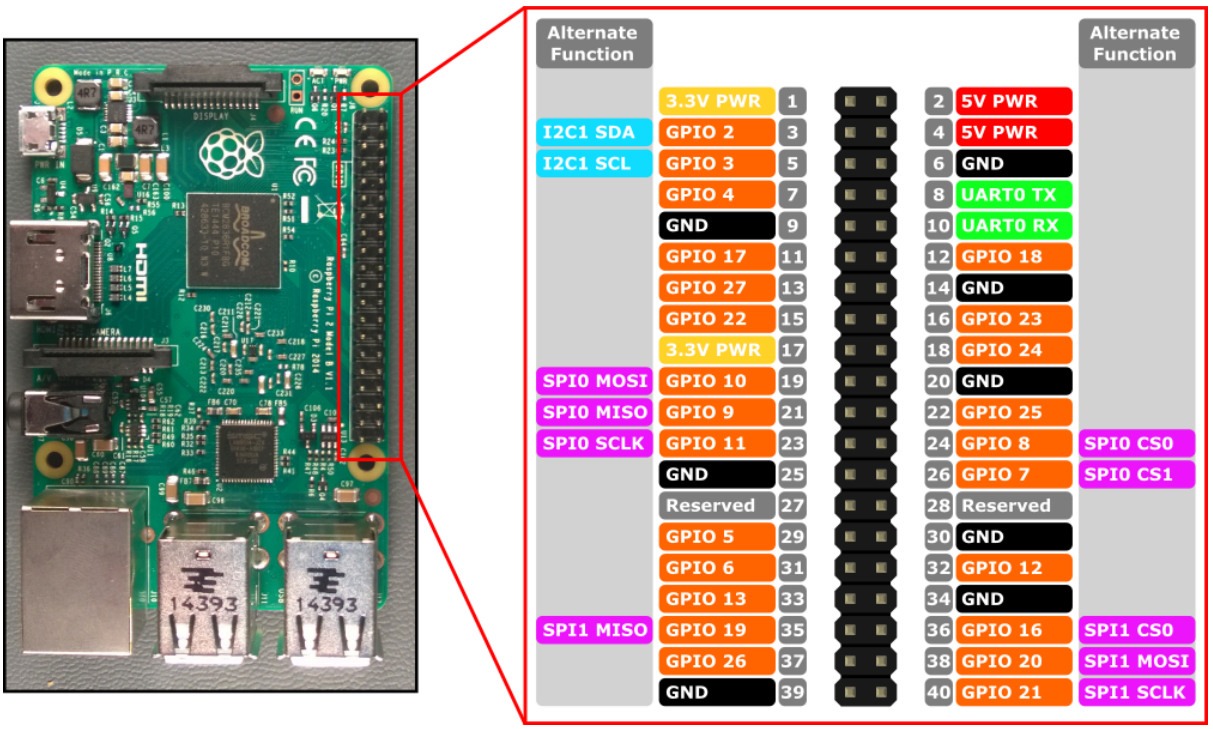
GPIO.setmode(GPIO.BCM)
ledPin = 22

GPIO.setup(ledPin, GPIO.OUT)
GPIO.output(ledPin, False)

try:
    while True:
        GPIO.output(ledPin, True)
        print("LED ON")
        sleep(1)

        GPIO.output(ledPin, False)
        print("LED OFF")
        sleep(1)

finally:
    GPIO.output(ledPin, False)
    GPIO.cleanup()
```


Alternate Function
Alternate Function
3.3V PWR 1
2 5V PWR
I2C1 SDA GPIO 2 3
4 5V PWR
I2C1 SCL GPIO 3 5
6 GND
GPIO 4 7
8 UART0 TX
GND 9
10 UART0 RX
GPIO 17 11
12 GPIO 18
GPIO 27 13
14 GND
GPIO 22 15
16 GPIO 23
3.3V PWR 17
18 GPIO 24
SPI0 MOSI GPIO 10 19
20 GND
SPI0 MISO GPIO 9 21
22 GPIO 25
SPI0 SCLK GPIO 11 23
24 GPIO 8 SPI0 CS0
GND 25
26 GPIO 7 SPI0 CS1
Reserved 27
28 Reserved
GPIO 5 29
30 GND
GPIO 6 31
32 GPIO 12
GPIO 13 33
34 GND
SPI1 MISO GPIO 19 35
36 GPIO 16 SPI1 CS0
GPIO 26 37
38 GPIO 20 SPI1 MOSI
GND 39
40 GPIO 21 SPI1 SCLK

Displaying different LED Patterns with Raspberry P

For displaying Differen LED pattern connecte 8 LEDs in the same format to the pin number given in the below python

Code (small code chatgpt vala) :

```
import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BOARD)

leds = [29, 31, 33, 35, 36, 37, 38, 40]

for led in leds:
    GPIO.setup(led, GPIO.OUT)
    GPIO.output(led, False)

def ledpattern(values):
    for led, value in zip(leds, values):
        GPIO.output(led, value)

def patternOne():
    for i in range(5):
        ledpattern([1, 0, 1, 0, 1, 0, 1, 0])
        time.sleep(1)
        ledpattern([0, 1, 0, 1, 0, 1, 0, 1])
        time.sleep(1)

def patternTwo():
    for i in range(5):
        ledpattern([1, 0, 0, 0, 0, 0, 0, 0])
        time.sleep(0.5)
        ledpattern([0, 1, 0, 0, 0, 0, 0, 0])
        time.sleep(0.5)
        ledpattern([0, 0, 1, 0, 0, 0, 0, 0])
        time.sleep(0.5)
        ledpattern([0, 0, 0, 1, 0, 0, 0, 0])
        time.sleep(0.5)

def patternThree():
    for i in range(5):
        ledpattern([1, 1, 1, 1, 1, 1, 1, 1])
        time.sleep(1)
        ledpattern([0, 0, 0, 0, 0, 0, 0, 0])
```

```
        time.sleep(1)

try:
    while True:
        patternOne()
        patternTwo()
        patternThree()

finally:
    GPIO.cleanup()
```

Code (big code clg vala) :

```
import RPi.GPIO as GPIO
import time

GPIO.setmode(GPIO.BOARD)

led1 = 29
led2 = 31
led3 = 33
led4 = 35
led5 = 36
led6 = 37
led7 = 38
led8 = 40

GPIO.setup(led1, GPIO.OUT)
GPIO.setup(led2, GPIO.OUT)
GPIO.setup(led3, GPIO.OUT)
GPIO.setup(led4, GPIO.OUT)
GPIO.setup(led5, GPIO.OUT)
GPIO.setup(led6, GPIO.OUT)
GPIO.setup(led7, GPIO.OUT)
GPIO.setup(led8, GPIO.OUT)

GPIO.output(led1, False)
GPIO.output(led2, False)
GPIO.output(led3, False)
GPIO.output(led4, False)
GPIO.output(led5, False)
GPIO.output(led6, False)
GPIO.output(led7, False)
GPIO.output(led8, False)

def ledpattern(ledVal1, ledVal2, ledVal3, ledVal4,
```

```
            ledVal5, ledVal6, ledVal7, ledVal8):
    GPIO.output(led1, ledVal1)
    GPIO.output(led2, ledVal2)
    GPIO.output(led3, ledVal3)
    GPIO.output(led4, ledVal4)
    GPIO.output(led5, ledVal5)
    GPIO.output(led6, ledVal6)
    GPIO.output(led7, ledVal7)
    GPIO.output(led8, ledVal8)

def patternOne():
    for i in range(5):
        ledpattern(1, 0, 1, 0, 1, 0, 1, 0)
        time.sleep(1)
        ledpattern(0, 1, 0, 1, 0, 1, 0, 1)
        time.sleep(1)

def patternTwo():
    for i in range(5):
        ledpattern(1, 0, 0, 0, 0, 0, 0, 0)
        time.sleep(0.1)
        ledpattern(0, 1, 0, 0, 0, 0, 0, 0)
        time.sleep(0.1)
        ledpattern(0, 0, 1, 0, 0, 0, 0, 0)
        time.sleep(0.1)
        ledpattern(0, 0, 0, 1, 0, 0, 0, 0)
        time.sleep(0.1)
        ledpattern(0, 0, 0, 0, 1, 0, 0, 0)
        time.sleep(0.1)
        ledpattern(0, 0, 0, 0, 0, 1, 0, 0)
        time.sleep(0.1)
        ledpattern(0, 0, 0, 0, 0, 0, 1, 0)
        time.sleep(0.1)
        ledpattern(0, 0, 0, 0, 0, 0, 0, 1)
        time.sleep(0.1)

def patternThree():
    for i in range(5):
        ledpattern(0, 0, 0, 0, 0, 0, 0, 1)
        time.sleep(0.1)
        ledpattern(0, 0, 0, 0, 0, 0, 1, 0)
        time.sleep(0.1)
        ledpattern(0, 0, 0, 0, 0, 1, 0, 0)
        time.sleep(0.1)
        ledpattern(0, 0, 0, 0, 1, 0, 0, 0)
        time.sleep(0.1)
        ledpattern(0, 0, 0, 1, 0, 0, 0, 0)
        time.sleep(0.1)
```

```
        ledpattern(0, 0, 1, 0, 0, 0, 0, 0)
        time.sleep(0.1)
        ledpattern(0, 1, 0, 0, 0, 0, 0, 0)
        time.sleep(0.1)
        ledpattern(1, 0, 0, 0, 0, 0, 0, 0)
        time.sleep(0.1)

try:
    while True:
        patternOne()
        patternTwo()
        patternThree()

finally:
    GPIO.cleanup()
```

Displaying Time over 4 Digit 7 Segment Display using Raspberry Pi

Hardware Guide:

For completing this lesson, you will require the following things along with your initial raspberry pi setup
1. TM1637 4-digit seven segment Display board
2. Connecting wires

Wiring up your Circuit:

Hook up your circuit as follows:
1. Connect the Pin2 (5V) of Rpi to Vcc pin of Module
2. Connect Pin 6 (GND) of Rpi to GND of Module
3. Connect Pin38 (GPIO20) of Rpi to DIO of Module
4. Lastly connect Pin 40 (GPIO21) of Rpi to CLK of Module

Software Guide (Short Notes)

1. Download the ZIP file from the https://github.com/timwaizenegger/raspberrypiexamples/tree/master/actor-led-7segment-4numbers
2. Find the downloaded file in `/home/pi/Downloads/`.
3. Extract the ZIP file and run the example codes in Python 2 IDLE.
4. Create a new Python file, write the given code, and save it in the same folder as `tm1637.py`.

Code :

```
from time import sleep

import tm1637


try:

    import thread

except ImportError:
```

```
    import _thread as thread

Display = tm1637.TM1637(CLK=21, DIO=20, brightness=1.0)

try:
    print("Starting clock in the background (press CTRL + C to stop):")
    Display.StartClock(military_time=True)
    Display.SetBrightness(1.0)

    while True:
        Display.ShowDoublepoint(True)
        sleep(1)
        Display.ShowDoublepoint(False)
        sleep(1)

except KeyboardInterrupt:
    print("Properly closing the clock and open GPIO pins")

finally:
    Display.StopClock()
    Display.cleanup()
```

Practical: Control LED Using Telegram Bot

Step 1: Install Telegram

Download and install Telegram on your phone.

Step 2: Create a Bot

1. Search for BotFather on Telegram.
2. Send `/newbot` and follow the instructions.
3. Copy the bot token and keep it secret.

Step 3: Install Telepot

Connect to Raspberry Pi and run:

sudo apt-get install python-pip

sudo pip install telepot

Step 4: Test the Bot Token

Run Python and enter:

import telepot

bot = telepot.Bot('YOUR_BOT_TOKEN')

bot.getMe()

Step 5: Python Program

Save the following code as `telegram.py`.

import time

import telepot

import RPi.GPIO as GPIO

from telepot.loop import MessageLoop

```
red = 40
GPIO.setmode(GPIO.BOARD)
GPIO.setwarnings(False)
GPIO.setup(red, GPIO.OUT)
GPIO.output(red, 0)

def action(msg):
    chat_id = msg['chat']['id']
    command = msg['text']

    if command == 'On':
        GPIO.output(red, 1)
        bot.sendMessage(chat_id, 'LED ON')

    if command == 'Off':
        GPIO.output(red, 0)
        bot.sendMessage(chat_id, 'LED OFF')

bot = telepot.Bot('YOUR_BOT_TOKEN')
MessageLoop(bot, action).run_as_thread()

while True:
    time.sleep(10)
```

Step 6: Connect LED

Connect the red LED to physical pin 40 (GPIO21) of the Raspberry Pi, with a suitable resistor and ground connection.

Step 7: Run the Program

Execute this command:

sudo python /home/pi/telegram.py

Open Telegram, search for your bot, and send On or Off to control the LED.

Camera Connection and capturing Images

Hardware Guide:
For completing this lesson, you will require the Camera Module along with your initial raspberry pi setup.

Camera Module:
The Raspberry Pi Camera Board plugs directly into the CSI connector on the Raspberry Pi.

The Raspberry Pi Camera Board Features:
1. Fully Compatible with Both the Model A and Model B Raspberry Pi
2. 5MP Omnivision 5647 Camera Module
3. Still Picture Resolution: 2592 x 1944
4. Video: Supports 1080p @ 30fps, 720p @ 60fps and 640x480p 60/90 Recording
5. 15-pin MIPI Camera Serial Interface – Plugs Directly into the Raspberry Pi Board
6. Size: 20 x 25 x 9mm
7. Weight 3g 8. Fully Compatible with many Raspberry Pi cases

Connect the Camera Module:
First of all, with the Pi switched off, you'll need to connect the Camera Module to the Raspberry Pi's camera port, then start up the Pi and ensure the software is enabled.

1. Locate the camera port and connect the camera:

Software Guide

Use these commands in the terminal to capture images and videos using the Raspberry Pi camera.

1. Capture an Image

sudo raspistill -o test.jpg

Captures an image and saves it as `test.jpg`.

2. Capture an Image with Delay and Resolution

libcamera-jpeg -o test1080.jpg -t 5000 --width 1920 --height 1080

Captures an image after a 5-second delay at 1920 × 1080 resolution.

3. Advanced Image Capture

- Save an image:

libcamera-still -o still-test.jpg

- Save as PNG:

libcamera-still -e png -o still-test.png

- Save with date and time:

libcamera-still --datetime

4. Capture Video

libcamera-vid -t 10000 -o test.h264

Records a 10-second video in H.264 format and saves it as `test.h264`.

—---------- over —----------